

$\{x_i\}_{i=1}^n \sim R(0, \theta), \theta > 0$, вер. модель
 $\theta \in \Theta = (0, +\infty)$
 выборка $\vec{x}_n$

$$\tilde{\theta}_1 = 2\bar{x} = 2 \frac{1}{n} \sum_{i=1}^n x_i$$

$$\tilde{\theta}_2 = x_{\min}$$

$$\tilde{\theta}_3 = x_{\max}$$

$$\tilde{\theta}_n = x_1 + \frac{1}{n-1} \sum_{i=2}^n x_i$$

$$M\zeta = \int_{-\infty}^{\infty} x dF(x, \theta) = \int_0^{\theta} x \frac{1}{\theta} dx = \frac{\theta}{2}$$

$$M\zeta^2 = \int_0^{\theta} x^2 \frac{1}{\theta} dx = \frac{\theta^2}{3}; \quad D\zeta = M\zeta^2 - (M\zeta)^2 = \frac{\theta^2}{12}$$

Параметрическая модель

Несмещённость: $\forall \theta \in \Theta \quad M\tilde{\theta} = \theta$

Состоятельность: $\forall \theta \in \Theta \quad \forall \varepsilon > 0 \quad P(|\tilde{\theta} - \theta| \geq \varepsilon) \xrightarrow{n \rightarrow \infty} 0$

$$① \tilde{\theta}_1 \quad \forall \theta > 0, \quad M\tilde{\theta}_1 = M\left(2 \frac{1}{n} \sum x_i\right) =$$

$$= \frac{2}{n} \sum M x_i = \frac{2}{n} n M\zeta = \theta \Rightarrow \text{несмещ.}$$

x_i незав. сл. вел. $x_i \sim R(0, \theta)$

$$D\tilde{\theta}_1 = D\left(2 \frac{1}{n} \sum x_i\right) = \frac{4}{n^2} \sum D x_i = \frac{4}{n^2} n D\zeta =$$

$$= \frac{\theta^2}{3n} \xrightarrow{n \rightarrow \infty} 0 \quad \Rightarrow \text{состоятельность по достат. условию} \quad \forall \theta > 0$$

$$\textcircled{2} \tilde{\Theta}_2 = x_{\min}$$

$$\forall \theta > 0, \mathcal{M} \tilde{\Theta}_2 = \mathcal{M} x_{\min}$$

$$z \sim F(x)$$

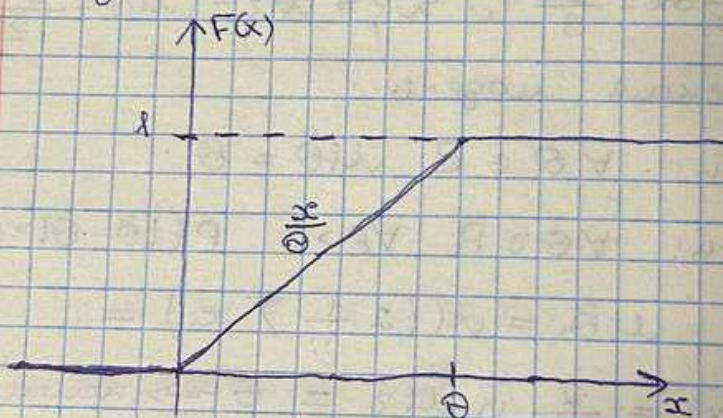
$$z_{\min} \sim \frac{1 - (1 - F(x))^n}{\varphi(x)}$$

$$\varphi(x) = \varphi'(x) = n (1 - F(x))^{n-1} F'(x) =$$

$$= n \left(1 - \frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} \mathbb{I}_{(0; \theta)}$$

$$\mathcal{M} x_{\min} = \int_0^{\theta} x n \left(1 - \frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} dx =$$

$$= \int_0^1 n(1-t) t^{n-1} \theta dt = \frac{\theta}{n+1} \Rightarrow \text{unbiased.}$$



$$\tilde{\Theta}_2' = (n+1) x_{\min}$$

$$\mathcal{M} \tilde{\Theta}_2' = (n+1)$$

$$\mathcal{M} \tilde{\Theta}_2' = (n+1) \mathcal{M} x_{\min} = \theta - \text{biased}$$

$$D\tilde{\theta}_2 = D((n+1)x_{\min}) = (n+1)^2 D x_{\min} = \frac{n\theta^2}{n+2} \xrightarrow{n \rightarrow \infty} 0$$

гос. уст. не поддается (?)

$$\begin{aligned} M x_{\min}^2 &= \int_0^{\theta} x^2 n \left(1 - \frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} dx = \\ &= \int_0^1 \theta^2 (1-t)^2 n t^{n-1} dt = n\theta^2 \int_0^1 (t^{n-1} - 2t^n + t^{n+1}) dt = \\ &= n\theta^2 \left(\frac{1}{n} - \frac{2}{n+1} + \frac{1}{n+2} \right) = \frac{\theta^2 \cdot 2}{(n+1)(n+2)} \end{aligned}$$

$$D x_{\min} = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \frac{n\theta^2}{(n+1)^2(n+2)}$$

$D\tilde{\theta}_2'$

$$\forall \theta > 0 \quad \forall \varepsilon > 0 \quad P(|\tilde{\theta}_2 - \theta| \geq \varepsilon) \xrightarrow{n \rightarrow \infty} 0$$



$$P(|\tilde{\theta}_2 - \theta| \geq \varepsilon) \geq P(\tilde{\theta}_2 \geq \theta + \varepsilon) =$$

$$= P((n+1)x_{\min} \geq \theta + \varepsilon) = P(x_{\min} \geq \frac{\theta + \varepsilon}{n+1}) = 1 - \Phi\left(\frac{\theta + \varepsilon}{n+1}\right)$$

$$\begin{aligned} \Phi(x) &= 1 - (1 - F(x))^n = 1 - \left(1 - \left(1 - \frac{\theta + \varepsilon}{\theta(n+1)}\right)^n\right) = \\ &= \left(1 - \frac{\theta + \varepsilon}{\theta(n+1)}\right)^n \rightarrow e^{-\frac{\theta + \varepsilon}{\theta}} > 0 \end{aligned}$$

$$\begin{aligned} P(|x_{\min} - \theta| \geq \varepsilon) &= P(x_{\min} \leq \theta - \varepsilon) = \Phi(\theta - \varepsilon) = \\ &= 1 - (1 - F(\theta - \varepsilon))^n = 1 - \left(1 - \frac{\theta - \varepsilon}{\theta}\right)^n = 1 - \left(\frac{\varepsilon}{\theta}\right)^n \xrightarrow{n \rightarrow \infty} 1 \end{aligned}$$

$$\exists \varepsilon > 0$$

$$\exists \theta > 0$$

не абн. гос.

$$\frac{\varepsilon}{\theta} < 1$$

$$\textcircled{3} \quad \tilde{\theta}_3 = x_{\max}$$

$$M\tilde{\theta}_3 = x_{\max}$$

$$x_{\max} \sim \frac{(F(x))^{n-1}}{f(x)}$$

$$f(x) = f'(x) = n \left(\frac{x}{\theta}\right)^{n-1} \cdot \frac{1}{\theta} \{0, \theta\}$$

$$Mx_{\max} = \int_0^{\theta} x n \left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} dx = \frac{n}{\theta^n} \cdot \frac{\theta^{n+1}}{n+1} =$$

$$= \frac{n}{n+1} \theta$$

$$\tilde{\theta}_3' = \frac{n+1}{n} x_{\max}$$

$$M\tilde{\theta}_3' = \frac{n+1}{n} Mx_{\max} = \theta - \text{несмещен}$$

$$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 Dx_{\max} = \frac{(n+1)^2}{n^2} \cdot \frac{n \theta^2}{(n+2)(n+1)^2} \textcircled{=}$$

$$Mx_{\max}^2 = \int_0^{\theta} x^2 n \left(\frac{x}{\theta}\right)^{n-1} \frac{1}{\theta} dx = \frac{n}{\theta^n} \frac{\theta^{n+2}}{n+2} =$$

$$= \frac{\theta^2 n}{n+2}$$

$$Dx_{\max} = \theta^2 \left(\frac{n}{n+2} - \frac{n^2}{(n+1)^2} \right) = \frac{n \theta^2}{(n+2)(n+1)^2}$$

$$\textcircled{=} \frac{\theta^2}{n(n+2)} \xrightarrow{n \rightarrow \infty} 0 \Rightarrow \text{состоит по глос. уст.} \forall \theta > 0$$

Док-во по определению.

$$1) \quad \tilde{\theta}_3: \quad \forall \epsilon > 0 \quad P(|\tilde{\theta}_3 - \theta| \geq \epsilon) \xrightarrow{n \rightarrow \infty} 0$$

$$P(|\tilde{\theta}_3 - \theta| \geq \epsilon) \geq P(\tilde{\theta}_3 \geq \epsilon + \theta) = P(x_{\max} \geq \theta + \epsilon) \neq$$

$$= 1 - F(\theta + \epsilon)$$

$$\Psi(x) = (F(x))^n =$$

$$P(|x_{\max} - \theta| \geq \varepsilon) = P(x_{\max} \leq \theta - \varepsilon) = \\ = \Psi(\theta - \varepsilon) = (F(\theta - \varepsilon))^n = \left(\frac{\theta - \varepsilon}{\theta}\right)^n = \left(1 - \frac{\varepsilon}{\theta}\right)^n \xrightarrow{n \rightarrow \infty} 0 \\ \Rightarrow \text{состоятельна}$$

$$2) \tilde{\theta}_3: \forall$$

$$P\left(\left|\frac{n+1}{n} x_{\max} - \theta\right| \geq \varepsilon\right) \geq P(\tilde{\theta}_3 \geq \theta + \varepsilon) = \\ = P\left(\frac{n+1}{n} x_{\max} \geq \theta + \varepsilon\right) = P\left(x_{\max} \geq \frac{n(\theta + \varepsilon)}{n+1}\right) = \\ = 1 - \Psi\left(\frac{n(\theta + \varepsilon)}{n+1}\right)$$

$$\Psi(x) = \Psi\left(\frac{n(\theta + \varepsilon)}{n+1}\right) = \left(F\left(\frac{n(\theta + \varepsilon)}{n+1}\right)\right)^n = \left(\frac{n(\theta + \varepsilon)}{\theta(n+1)}\right)^n \xrightarrow{n \rightarrow \infty} 1 \\ \Rightarrow P(|\tilde{\theta}_3 - \theta| \geq \varepsilon) \xrightarrow{n \rightarrow \infty} 0 \\ \Rightarrow \text{состоятельна}$$

$$4) \tilde{\theta}_4 = x_1 + \frac{1}{(n-1)} \sum_{i=2}^n x_i$$

$$M\tilde{\theta}_4 = Mx_1 + \frac{1}{n-1} \sum_{i=2}^n Mx_i = Mx_1 + \frac{1}{n-1} \sum_{i=2}^n Mx_i =$$

$$= \theta \Rightarrow \text{несмещ.} \\ D\tilde{\theta}_4 = Dx_1 + \frac{1}{(n-1)^2} \sum_{i=2}^n Dx_i = \frac{\theta^2}{12} + \frac{1}{(n-1)} \frac{\theta^2}{12} \not\rightarrow 0 \\ \tilde{\theta}_4 \xrightarrow{P} \theta \quad \text{не вып. дост.} \\ \text{усл.}$$

$$x_1 + \frac{1}{n-1} \sum_{i=2}^n x_i \xrightarrow{P} \theta + \frac{\theta}{2} \quad \text{не явл. сост.}$$

$$\begin{matrix} \{z_n\} \xrightarrow{P} \{z\} \\ \{\eta_n\} \xrightarrow{P} \{\eta\} \end{matrix} \rightarrow \begin{matrix} \{z_n + \eta_n\} \xrightarrow{P} \{z + \eta\} \\ \{z_n \cdot \eta_n\} \rightarrow \{z \cdot \eta\} \end{matrix}$$

ЗБЧ Хинчина $\{z_n\}$ незав.
 един. распр., $\exists Mz_n < \infty$
 $\rightarrow \frac{1}{n} \sum_{i=1}^n z_n \rightarrow Mz = \frac{n\theta^2}{(n+2)(n+1)^2}$

$$\tilde{\theta}_3 = \frac{n+1}{n} x_{\max}$$

$$\tilde{\theta}_1 = 2\bar{x}$$

$$D\tilde{\theta}_1 = \frac{\theta^2}{3n}$$

$$D\tilde{\theta}_3 = \frac{\theta^2}{(n+2)(n+1)}$$

$$\frac{1}{3n} \quad \frac{1}{n(n+2)}$$

$$n^2 + 2n > 3n \Rightarrow \tilde{\theta}_3 \text{ более точ. } \tilde{\theta}_1 \\ n > 1$$